Code No: 114CS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, September/October - 2023
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	What is time complexity?	[2]
b)	Differentiate between Pseudocode and an algorithm.	[3]
c)	Write an algorithm of Union operation.	[2]
d)	What is the importance of Game tree?	[3]
e)	What is feasible solution?	[2]
f)	Write an algorithm of Greedy Knapsack.	[3]
g)	What is state space tree?	[2]
h)	What are the advantages of Backtracking?	[3]
i)	What is Non-Polynomial time algorithm?	[2]
j)	What is decision problem?	[3]

PART - B

(50 Marks)

2.a) Derive the time complexity of Strassen's matrix multiplication.
b) Explain about the Amortized complexity. [5+5]

OR

3. Write and explain the algorithm of Merge Sort and also analyze the time complexity of the same. [10]

4. Write an algorithm to find the connected components using DFS and also analyze the time complexity of the same. [10]

OR

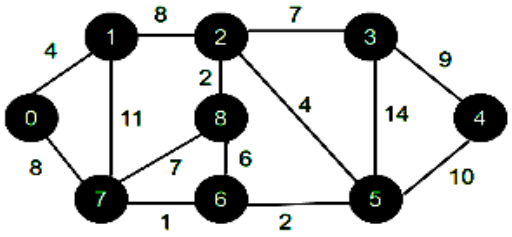
5.a) Explain about the AND/OR graph with an example.
b) Write an algorithm of BFS and also analyze the time complexity of the same. [5+5]

6. Design a three stage system with device types D1, D2, D3. The costs are $30, $15, $20 respectively. The cost of the system is to be no more than $105. The reliability of each device type is 00.9, 0.8 and 0.5 respectively. [10]

OR

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7.a) Find the shortest path of the following graph from source to destination by using single source shortest path.



b) Write down the differences between prim's and Krushkal's algorithm. [6+4]

8.a) Draw the portion of the state space tree generated by LCBB for the following knapsack instances:

n=5, (P1, P2, P3, P4, P5) = (10, 15, 6, 8, 4)

(W1, W2, W3, W4, W5) = (4, 6, 3, 4, 2) and m = 12

b) Write down the applications of n-Queens problem. [6+4]

OR

9.a) Explain how to find the Hamiltonian cycle using Backtracking.

b) Draw and explain the tree organization of the 4 – Queen Solution state space tree. [5+5]

10.a) Write a non-deterministic algorithm of Sorting of an elements in an array.

b) Explain about the Cook's theorem. [5+5]

OR

11.a) Explain the relationship between P, NP, NP-Hard and NP-Complete.

b) Write a Non – deterministic algorithm of 0/1Knapsack problem. [5+5]

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